

Notes: Mortensen and Pissarides (RES, 1994)

Job Creation and Job Destruction in the Theory of Unemployment

Contents

1	Big picture	3
2	One-sentence summary	3
3	Incremental contribution of the paper	3
4	Economic environment and model primitives	4
4.1	Jobs, workers, and productivities	4
4.2	Creation of new jobs	4
4.3	Wage determination and surplus sharing	5
5	Core equilibrium conditions	5
5.1	Vacancy value and free entry	5
5.2	Filled-job value and destruction threshold	5
5.3	Job destruction condition	6
5.4	Job creation condition	6
5.5	Beveridge curve and steady-state unemployment	6
6	Steady-state comparative statics	7
6.1	Aggregate productivity shocks	7
6.2	Dispersion shocks	7
6.3	Job-specific shock frequency	7
7	Cyclical dynamics	8
7.1	Aggregate productivity as a Poisson process	8
7.2	Why job destruction is more volatile than job creation	8
8	Simulation and quantitative comparison	8
8.1	Calibration strategy	8
8.2	Simulation results	9
9	Identification and interpretation	9

10	Extracted figures and tables from the paper	10
10.1	Figure 1. The joint determination of v/u and ε_d	10
10.2	Figure 2. Equilibrium vacancies and unemployment	10
10.3	Table I. Parameter values	11
10.4	Table II. Simulation results: Means (std errors) of 100 simulated 66-quarter samples	11
11	Short summary	11
12	Conclusion	12
12.1	Core conclusion	12
12.2	Incremental contribution revisited	12
12.3	Limitations	12
12.4	Takeaway	12

1 Big picture

This paper builds a search-and-matching model in which unemployment dynamics are driven not only by job creation through matching, but also by endogenous job destruction. The key innovation is that existing jobs are hit by persistent job-specific shocks, and firms optimally decide whether a low-productivity job should continue or be destroyed. The model is therefore able to speak directly to the empirical fact that large job creation and job destruction flows coexist at every phase of the business cycle.

The central idea is simple but powerful. New jobs are created at the best available productivity, because firms open vacancies only when expected surplus is high enough. Existing jobs, however, are subject to idiosyncratic productivity shocks. If a job receives a sufficiently bad shock, the continuation surplus becomes negative and the job is destroyed. The economy therefore has two endogenous margins: a job-creation margin governed by free entry of vacancies, and a job-destruction margin governed by a reservation productivity threshold.

The paper's main message is that aggregate shocks and dispersion shocks have different implications for job flows. A positive aggregate productivity shock raises job creation and lowers job destruction, producing negative co-movement between the two flows. By contrast, a dispersion shock raises both job creation and job destruction, producing positive co-movement. This distinction makes job-flow data informative about the nature of shocks driving unemployment fluctuations.

2 One-sentence summary

This note summarizes Mortensen and Pissarides (1994), which embeds job-specific productivity shocks into a matching model with wage bargaining, derives endogenous job creation and job destruction through a vacancy free-entry condition and a reservation-productivity destruction rule, and shows that aggregate productivity shocks can reproduce the negative correlation and relative volatility of U.S. job creation and job destruction.

3 Incremental contribution of the paper

The paper contributes to the unemployment and matching literature in four main ways.

1. **Endogenous job destruction inside a matching model.** Earlier matching models often treat job separation as exogenous. This paper endogenizes separation by allowing firms to close filled jobs when job-specific productivity falls below a reservation threshold.
2. **A joint theory of job creation and job destruction.** The model does not study hiring alone. It explains simultaneous gross job creation and gross job destruction, which were emphasized in the empirical work of Davis and Haltiwanger.
3. **A distinction between aggregate and reallocation shocks.** Aggregate shocks move job creation and job destruction in opposite directions, whereas dispersion shocks move them in the same direction. This gives the theory sharper empirical content.
4. **A dynamic explanation for asymmetric unemployment movements.** Job destruction reacts sharply on impact when aggregate productivity falls, because all jobs between the old and new destruction thresholds are destroyed immediately. Job creation adjusts through vacancies

and matching, making it smoother. This generates more volatile job destruction than job creation.

The paper's deeper contribution is methodological. It shows that stock-flow unemployment models should not focus only on the Beveridge curve or unemployment-vacancy stocks. Gross job-flow data contain extra information about the underlying shocks because job creation and job destruction respond differently to aggregate and idiosyncratic forces.

4 Economic environment and model primitives

4.1 Jobs, workers, and productivities

Each firm has one job. A job can be either filled and producing, vacant and searching, or destroyed. Workers can be employed or unemployed and searching. There is no on-the-job search, so worker flows and job flows coincide in the model.

A filled job produces one unit of a differentiated product. The value of output is

$$p + \sigma\varepsilon,$$

where:

- p is the aggregate component of productivity, common to all jobs;
- ε is a job-specific component;
- σ measures dispersion in job-specific productivity.

The idiosyncratic component follows a Poisson jump process. With arrival rate λ , a job receives a new draw from distribution $F(\varepsilon)$ with finite upper support ε_u . A sufficiently bad draw causes job destruction.

Interpretation. The model separates common aggregate productivity movements from job-specific shocks. Aggregate shocks shift all jobs together; idiosyncratic shocks reshuffle the relative profitability of existing jobs.

4.2 Creation of new jobs

New jobs are assumed to enter at the upper support of the productivity distribution, $p + \sigma\varepsilon_u$. This captures the idea that firms opening new vacancies choose the best currently available product or technology. Once created, however, a job is irreversible and cannot switch product varieties after receiving bad shocks.

Vacancies cost c per unit time. A vacancy meets an unemployed worker through a constant-returns matching function,

$$m(v, u),$$

where v is vacancies and u is unemployment. The vacancy filling rate is

$$q\left(\frac{v}{u}\right) = \frac{m(v, u)}{u} = m\left(1, \frac{v}{u}\right),$$

and the worker job-finding rate is

$$\frac{vq(v/u)}{u} = m\left(\frac{v}{u}, 1\right).$$

Interpretation. Market tightness v/u summarizes the congestion structure of the search market. A higher v/u means that unemployed workers find jobs more easily, but vacancies are harder to fill.

4.3 Wage determination and surplus sharing

Wages are set by bilateral bargaining and continuously revised after productivity shocks. Let $J(\varepsilon)$ be the asset value of a filled job to the firm, $W(\varepsilon)$ the worker's value of employment, and U the value of unemployment. Match surplus is

$$S(\varepsilon) = J(\varepsilon) + W(\varepsilon) - U.$$

The worker receives share β of the surplus:

$$W(\varepsilon) - U = \beta S(\varepsilon),$$

and the firm receives $(1 - \beta)S(\varepsilon)$.

Interpretation. The wage rule links labour-market tightness to the opportunity cost of employment. When vacancies are abundant, workers' outside options rise, raising the threshold at which jobs are worth keeping.

5 Core equilibrium conditions

5.1 Vacancy value and free entry

The asset value of a vacancy satisfies

$$rV = -c + q(v/u)[J(\varepsilon_u) - V].$$

Free entry of vacancies drives vacancy value to zero:

$$rV = 0.$$

Therefore, the cost of maintaining a vacancy equals the expected capital gain from meeting a worker and forming a productive job.

5.2 Filled-job value and destruction threshold

The firm can destroy a job at no cost. Therefore a job survives only if the firm's value is nonnegative:

$$J(\varepsilon) = (1 - \beta)S(\varepsilon) \geq 0.$$

Since surplus is increasing in ε , there is a unique reservation productivity ε_d such that

$$S(\varepsilon_d) = 0.$$

Jobs with $\varepsilon < \varepsilon_d$ are destroyed after a shock.

The endogenous job destruction rate is therefore

$$\lambda F(\varepsilon_d).$$

Interpretation. Job destruction is not imposed from outside the model. It is the optimal stopping rule of firms facing irreversible jobs and stochastic idiosyncratic productivities.

5.3 Job destruction condition

The reservation productivity condition is

$$p + \sigma \varepsilon_d = b + \frac{\beta c}{1 - \beta} \frac{v}{u} - \frac{\sigma \lambda}{r + \lambda} \int_{\varepsilon_d}^{\varepsilon_u} [1 - F(x)] dx.$$

This equation equates the output value of the marginal surviving job to three components:

- b : the value of unemployment or leisure;
- $\frac{\beta c}{1 - \beta} \frac{v}{u}$: the worker's search opportunity cost, rising in market tightness;
- $-\frac{\sigma \lambda}{r + \lambda} \int_{\varepsilon_d}^{\varepsilon_u} [1 - F(x)] dx$: the option value of keeping a currently low-productivity job alive in case future idiosyncratic shocks improve its value.

Economic message. The firm may keep a job even when current productivity is low because the job has option value. This is the model's labour-hoarding mechanism.

5.4 Job creation condition

The free-entry condition for vacancies implies

$$q\left(\frac{v}{u}\right) = \frac{c}{1 - \beta} \frac{r + \lambda}{\sigma(\varepsilon_u - \varepsilon_d)}.$$

A higher destruction threshold ε_d lowers the surplus of new jobs because the upper-support job has less distance from the destruction point. Therefore, higher expected destruction reduces job creation.

Economic message. Job creation and job destruction are linked. Vacancy entry depends on the surplus of newly created jobs, and that surplus depends on how easily jobs become unprofitable and are destroyed.

5.5 Beveridge curve and steady-state unemployment

In steady state, flows into unemployment equal flows out of unemployment. The unemployment rate satisfies

$$u = \frac{\lambda F(\varepsilon_d)}{\lambda F(\varepsilon_d) + m(v/u, 1)}.$$

This is the model's Beveridge-curve relation. Unlike in a standard matching model with exogenous separation, here the separation rate itself depends on market tightness through ε_d .

6 Steady-state comparative statics

6.1 Aggregate productivity shocks

A positive aggregate productivity shock, represented by a higher p , increases the value of all jobs. It lowers the destruction threshold ε_d and increases market tightness v/u .

Consequences:

- job creation rises because vacancies are more profitable;
- job destruction falls because fewer existing jobs fall below the reservation threshold;
- unemployment falls in the new steady state;
- the effect on vacancies is ambiguous because unemployment falls while tightness rises.

Economic message. Aggregate productivity shocks generate negative co-movement between job creation and job destruction. In booms, firms create more jobs and destroy fewer jobs; in recessions, they create fewer jobs and destroy more jobs.

6.2 Dispersion shocks

An increase in σ raises the dispersion of idiosyncratic productivity. New jobs, created at the upper support $p + \sigma\varepsilon_u$, become more profitable, while low-productivity existing jobs become worse. Therefore:

- job creation rises because new jobs are more productive;
- job destruction rises because low-productivity jobs are more likely to fall below the reservation threshold;
- vacancies rise;
- the effect on unemployment is ambiguous.

Economic message. Reallocation shocks create positive co-movement between job creation and job destruction. They increase gross flows in both directions rather than generating a simple boom-bust pattern.

6.3 Job-specific shock frequency

An increase in λ makes idiosyncratic shocks arrive more often. This has two opposing effects on destruction:

- mechanically, shocks arrive more often;
- but because bad jobs are less persistent, the option value of waiting rises, reducing the reservation threshold.

The paper shows that higher λ lowers job creation because new jobs face less persistent productivity states and therefore lower continuation surplus.

7 Cyclical dynamics

7.1 Aggregate productivity as a Poisson process

The paper extends the model by allowing aggregate productivity p to switch between a high value p^* and a low value p according to a Poisson process. This creates a dynamic difference between jump variables and state variables:

- market tightness v/u and the destruction threshold ε_d jump when aggregate productivity changes;
- unemployment adjusts gradually through matching and destruction flows.

When aggregate productivity falls from p^* to p , vacancies close down and all jobs whose idiosyncratic productivity lies between the old and new reservation thresholds are destroyed immediately. This creates a sharp spike in job destruction at the start of a recession.

Economic message. Recessions are characterized by a fast destruction margin and a slower creation margin. The immediate jump in the destruction threshold makes job destruction more volatile than job creation.

7.2 Why job destruction is more volatile than job creation

The model generates asymmetric dynamics because job destruction has an impact effect while job creation works through vacancy posting and matching. When productivity falls:

- job creation falls on impact because vacancies are less profitable;
- job destruction rises on impact because marginal jobs are immediately destroyed;
- unemployment rises faster at the start of recessions than it falls at the start of booms.

This mechanism rationalizes why, in the data, job destruction is more volatile and leads unemployment increases during downturns.

8 Simulation and quantitative comparison

8.1 Calibration strategy

The simulation calibrates a three-state Markov chain for aggregate productivity to match quarterly deviations of log labour productivity from a linear trend in U.S. manufacturing from 1947 to 1991. The idiosyncratic component is assumed uniform on $[-1, 1]$.

The matching function is log-linear, and the worker meeting rate is written as

$$a(p) = k[S(\varepsilon_u, p)]^{\theta/(1-\theta)}.$$

The remaining parameters are chosen to match the mean and volatility of U.S. manufacturing unemployment and the mean and standard deviation of job creation reported by Davis and Haltiwanger.

8.2 Simulation results

The simulation produces:

- mean job creation close to the U.S. manufacturing data;
- job destruction more volatile than job creation;
- negative correlation between creation and destruction;
- negative correlation between vacancies and unemployment, consistent with a Beveridge curve.

The simulated correlation between creation and destruction is less negative than the data, but the data statistic lies within one standard error of the simulation mean. The model therefore captures the broad qualitative and quantitative job-flow patterns using aggregate productivity shocks alone.

9 Identification and interpretation

The paper's theoretical identification is based on the distinct signs of job-flow responses:

- aggregate productivity shocks: job creation up, job destruction down;
- dispersion/reallocation shocks: job creation up, job destruction up.

Therefore, the covariance between job creation and job destruction provides information about the type of shock driving unemployment. Negative co-movement is evidence for aggregate productivity shocks; positive co-movement would be more consistent with reallocation shocks.

This is why the paper argues that vacancy and unemployment stocks alone are not sufficient. Flow data on job creation and destruction are more informative because they reveal whether the labour market is expanding, contracting, or reallocating.

10 Extracted figures and tables from the paper

10.1 Figure 1. The joint determination of v/u and ε_d

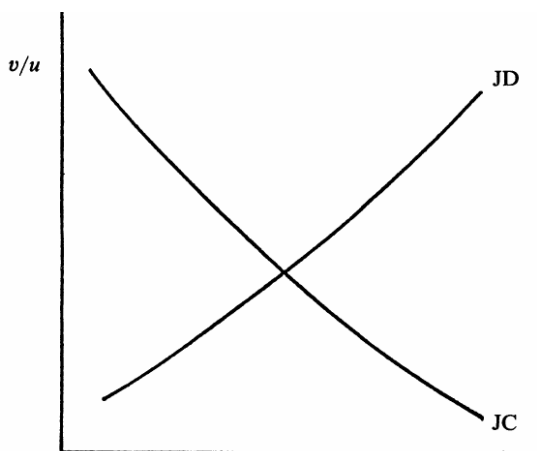


FIGURE 1
The joint determination of v/u and ε_d

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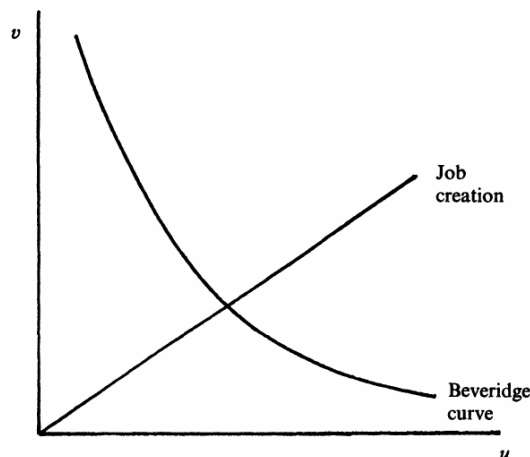


FIGURE 2
Equilibrium vacancies and unemployment

Figure 2. Equilibrium vacancies and unemployment

Read: Figure 1 plots the two core equilibrium loci. The upward-sloping JD curve is the job-destruction condition: higher market tightness increases workers' outside options, raising the reservation productivity. The downward-sloping JC curve is the job-creation condition: higher reservation productivity lowers the surplus of new jobs, reducing vacancy creation.

Economic message: Job creation and job destruction are jointly determined. A shock that shifts either locus changes both vacancy creation and the destruction threshold.

10.2 Figure 2. Equilibrium vacancies and unemployment

Read: Figure 2 maps the equilibrium into vacancy-unemployment space. The Beveridge curve equates unemployment inflows and outflows, while the job-creation line pins down the equilibrium market-tightness ratio.

Economic message: With endogenous job destruction, shocks can shift both the job-creation line and the Beveridge curve. This is why unemployment-vacancy stocks alone cannot fully distinguish aggregate shocks from reallocation shocks.

10.3 Table I. Parameter values

TABLE I Parameter values	
$\eta - b = 0.075$ Average net productivity	$r = 0.01$ Pure discount rate
$\lambda = 0.081$ Job-specific shock arrival rate	$\beta = 0.5$ Worker's share of surplus
$\sigma = 0.0375$ Job-specific shock dispersion	$\theta = 0.5$ Search elasticity of matching
	$k = 4$ Matching rate scale parameter

Table I. Parameter values

TABLE II
Simulation results: Means (std errors) of 100 Simulated 66 Quarters Samples

	Simulation statistics	Data
Mean (<i>c</i>)	5.23 (0.37)	5.2 ¹
Std. Dev. (<i>c</i>)	0.91 (0.43)	0.9 ¹
Std. Dev. (<i>d</i>)	1.37 (0.68)	1.6 ¹
Corr (<i>c</i> , <i>d</i>)	-0.10 (0.28)	-0.36 ¹

Table II. Simulation results: Means (std errors) of 100 simulated 66-quarter samples

Read: Table I reports the parameter choices for the simulation, including average net productivity, the arrival rate and dispersion of job-specific shocks, the discount rate, the worker's surplus share, the search elasticity of matching, and the matching-rate scale parameter.

Economic message: The calibration is deliberately parsimonious. The model tries to explain job-flow dynamics with a small set of search, bargaining, discounting, and idiosyncratic-shock parameters.

10.4 Table II. Simulation results: Means (std errors) of 100 simulated 66-quarter samples

Read: Table II compares simulated job creation and job destruction statistics with U.S. manufacturing job-flow data. The simulation matches the mean and standard deviation of job creation by construction and produces job destruction that is more volatile than job creation. It also produces a negative creation-destruction correlation and a negative vacancy-unemployment correlation.

Economic message: The calibrated model can reproduce the central qualitative patterns of job-flow data: creation and destruction are negatively correlated over the cycle, and destruction is more volatile.

11 Short summary

The paper introduces job-specific productivity shocks into a matching model of unemployment. Vacancies are created at the best available productivity, but filled jobs are exposed to idiosyncratic shocks. Firms destroy jobs whose productivity falls below an endogenous reservation threshold. The result is a theory of both job creation and job destruction.

Aggregate productivity shocks increase job creation and reduce job destruction, producing negative co-movement. Dispersion shocks increase both creation and destruction, producing positive co-movement. This distinction is the paper's key empirical insight. Simulations show that aggregate shocks alone can approximate the observed U.S. manufacturing patterns: job destruction is more volatile than job creation, and the two flows are negatively correlated.

12 Conclusion

12.1 Core conclusion

The core conclusion is that unemployment dynamics cannot be understood only through net employment changes. Gross job creation and job destruction contain separate information about how firms react to aggregate and idiosyncratic shocks. Endogenous job destruction is essential for understanding the cyclical dynamics of unemployment.

12.2 Incremental contribution revisited

The paper’s incremental contribution is to transform the search-and-matching model from a theory of vacancy creation into a theory of labour-market reallocation. The destruction margin makes the model consistent with establishment-level evidence showing simultaneous job creation and job destruction, and it gives the model richer cyclical implications.

12.3 Limitations

Several simplifying assumptions are important:

- each firm has one job, so worker flows and job flows coincide;
- there is no on-the-job search;
- new jobs enter at the upper support of the productivity distribution;
- job creation and destruction are costless, while vacancy posting is costly;
- the simulation uses a parsimonious aggregate productivity process.

These assumptions make the model transparent, but later work extends the framework to richer firm dynamics, on-the-job search, endogenous productivity choices, and richer aggregate shocks.

12.4 Takeaway

The paper is foundational because it explains why recessions involve both fewer new jobs and more job destruction, and why job destruction is especially volatile. It also shows that job-flow data help identify the underlying source of labour-market shocks. In this sense, the Mortensen-Pissarides framework is not only a theory of unemployment, but also a theory of labour-market reallocation.